

# Ananya Penumudi

EMBEDDED & IOT ENGINEER

B.TECH CSE (IOT) · CLASS OF 2027

B.Tech CSE (IoT) student, graduating in **2027**, building **embedded systems, AI-powered devices and safety-critical software**. Industrial trainee at **BHEL Bengaluru** and author of a systems-engineering thesis on **KAVACH**, India's Automatic Train Protection system.



## PORTFOLIO

[ananyapenumudi-ops.github.io]

## EMAIL

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## LINKEDIN

[in/ananya-penumudi]

## GITHUB

[github.com/ananyapenumudi-ops]

## EDUCATION

### VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING & TECHNOLOGY

2023 – 2027 (expected)

B.Tech, Computer Science & Engineering (IoT) · JNTUH · CGPA 8.92 / 10 (till Semester VI)

### SRI CHAITANYA JUNIOR COLLEGE

2021 – 2023

Intermediate, Telangana State Board · 97.4%

### BHARATIYA VIDYA BHAVAN'S PUBLIC SCHOOL

2021

High School, CBSE · GPA 9.38 / 10

## WORK EXPERIENCE

### BHARAT HEAVY ELECTRICALS LIMITED (BHEL), BENGALURU

Jun – Jul 2026

Industrial Trainee → Internship certificate

- Trained at the Solar Business Division on manufacturing, production workflows, quality assurance and system integration in a Government of India PSU.
- Studied railway signalling and KAVACH through technical discussions and documentation.

### ALFAGO

Jan 2026

Software Engineering Intern · Position Tracker → Internship certificate

- Built tax-lot based profit-and-loss calculations and backend transaction logic for a tracker covering transaction- and portfolio-level PnL.

### KARTGEN INFOTECH LLP

Jan – Apr 2025

Operations Intern (remote) → Letter of recommendation

- Supported business development and operations: process coordination and stakeholder communication, under Shrivatsa Malviya.

## FEATURED PROJECTS

### ELEMENTIUM AI: VIRTUAL CHEMISTRY LAB

2026

React Three Fiber · FastAPI · MongoDB Atlas · Gemini → GitHub

- Built a 360° 3D lab with drag-and-drop apparatus, real-time titration and colour-change chemistry.
- Added a Gemini-powered tutor and a rule-based engine that catches unsafe or out-of-order steps; the lab notebook fills itself in.

## BEYOND COLLISION PREVENTION: KAVACH THESIS

Jun 2026

Systems-engineering research · BHEL guidance → Thesis report → Simulator → Digital twin

- Analysed the architecture, operating modes, communication framework and functional safety of India's Automatic Train Protection system.
- Built an interactive simulator of KAVACH's modes of operation and a digital twin.

## MORE PROJECTS

### SMART AI WHEELCHAIR SYSTEM

Jan 2026

IoT + edge AI → Technical report

- Vision-based obstacle detection with automatic braking; GPS geofencing, caregiver alerts and sensor fusion across camera, IMU and proximity sensors.

### AI-ENHANCED SECURE DATA WIPING PLATFORM

Nov 2025

AI + cybersecurity · team project → GitHub

- Binary-overwrite data sanitisation with cryptographic verification of every wipe and AI-based anomaly detection.

### SYNCOPE: FALL DETECTION & ALERT SYSTEM

Aug 2025

Wearable IoT → Documentation

- A wearable that detects falls from motion-sensor data and alerts caregivers automatically.

### FOCUSFLOW: ADHD-FRIENDLY PRODUCTIVITY PLATFORM

Apr 2025

AI + web → GitHub

- Mood tracking, journaling, facial-expression analysis and gamified focus tools.

### AURALYN: EMOTIONAL-WELLNESS WEARABLE

2025 – now

IoT + AI + healthcare · in design

- Designing an ESP32-S3 + BLE wearable that pairs biometric sensing with mood analysis, with apps for the wearer, therapist and family.

### EARLIER WEB PROJECTS

2023 – 2024

- **Ideavoid**: startup-idea validation platform with peer feedback and market-viability assessment.
- **Cheugy**: wellness app for stress management and self-care. → GitHub
- **Syndicate**: gamified learning platform built around films, documentaries, quizzes and discussions.

## TECHNICAL SKILLS

### IOT & EMBEDDED

Arduino, ESP32, sensor integration (SPI, I2C, PWM), BLE, GPS, GSM, accelerometers

### PROGRAMMING

C, C++, Python, Java, JavaScript, R, HTML, CSS, data structures & algorithms

### AI & DATA

Computer vision, edge inference, mood detection, anomaly detection, Tableau

### SYSTEMS ENGINEERING

Requirements analysis, functional safety, verification & validation, system architecture, traceability

### WEB, CLOUD & TOOLS

React Three Fiber, FastAPI, MongoDB, MySQL, AWS, CI/CD, Figma, AutoCAD, StarUML

### LANGUAGES

English, Hindi, Telugu

## CERTIFICATIONS

- **DevOps with Real-Time Practical Exposure**, Brainovision (AICTE approved), VNRVJIET · Jul – Nov 2025
- **Tableau Certified Data Analyst Training**, Udemy (59 hours) · Jun 2024 · → Certificate
- **RoboJAM Robotics Workshop**, Convergence 2K23, VNRVJIET · Dec 2023

## AWARDS & ACTIVITIES

- **2nd Prize, IDEATHON 2023**, VNRVJIET · → Certificate
- **Smart India Hackathon 2025**: solution design, prototyping and technical documentation
- 24-hour Hackathon, Convergence 2025, VNRVJIET · → Certificate
- 24-hour Hackathon DEMUX 2024, BVRIT · → Certificate
- National Conclave 2025, CBIT · → Certificate
- Member, Computer Society of India (VNRVJIET chapter) and Art of Living Cultural Club, 2023 – present